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Gorlos-phos: addressing the stereoselectivity in palladium-catalyzed *exo*-mode cyclization of allenes with a nucleophilic functionality†

Juntao Ye,[a] Suhua Li[a] and Shengming Ma*[a,b]

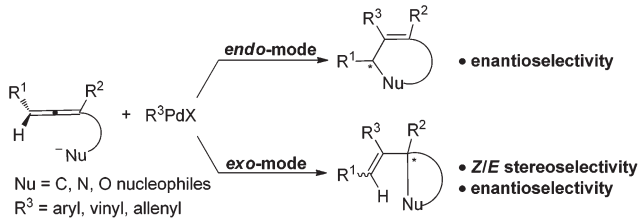
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A novel catalyst system has been indentified for addressing the long-standing issue of *Z*/*E* stereoselectivity in palladium-catalyzed *exo*-mode cyclization reactions of allenes bearing a nucleophilic functionality with organic halides or their equivalents. The readily accessible, sterically hindered monophosphine ligand **Gorlos-Phos·HBF$_4$** imparts a remarkable stereocontrolling ability with broad generality under Pd catalysis.

One of the long-standing and major challenges that organic chemists face is how to control the chemo-, regio- and stereoselectivity of a chemical reaction.[1] Allenes are a class of compounds endowed with a unique reactivity due to the presence of two consecutive C=C bonds, both of which are highly reactive under appropriate conditions and thus provide opportunities for multiple elaborations; however, they also pose a great challenge to control the selectivity in their reactions.[2] Among the extensive transformations of allenes, palladium-catalyzed cyclization reactions of allenes bearing a nucleophilic moiety with organic halides or their equivalents represent one of the most powerful tools for constructing a wide variety of carbo- and heterocycles *via endo*- or *exo*-mode cyclization reactions (Scheme 1).[3] While much attention has been paid to the enantioselectivity issue in both cases for the synthesis of optically active compounds *via* an axial-to-central chirality transfer process,[3b] control of the *Z*/*E* stereoselectivity referred to the *exo*-mode cyclization ($R^1 \neq H$) remains a challenge (Scheme 1),[4,5] primarily owing to the enormous difficulty in identifying a catalyst system having efficiency in both reactivity and stereoselectivity. Herein, we report a highly efficient and reliable strategy to solve such a challenging issue of *Z*/*E* stereoselectivity by taking advantage of the readily accessible, sterically hindered monophosphine ligand **Gorlos-Phos·HBF$_4$**.[6a]



Scheme 1 Cyclization modes of allenes with a nucleophilic functionality.

The reaction of 4-monosubstituted 2,3-allenyl amine **1a** with propargylic carbonate **2a** afforded the corresponding cyclization product **3aa** as a mixture of *Z*/*E* stereoisomers with a ratio of *Z*/*E* = 82 : 18 (Table 1, entry 1).[4d] Thus, we started to develop a new protocol addressing the issue of *Z*/*E*-selectivity by choosing an appropriate combination of palladium catalyst and phosphine ligand with different electronic and/or steric nature (Table 1). When different palladium catalysts (Table 1, entries 2–5) were used in combination with TFP (TFP = tri(2′-furyl)phosphine), $Pd_2(dba)_3{\cdot}CHCl_3$ gave the best results in terms of both the yield and stereoselectivity of **3aa** (Table 1, entry 5). The electron-rich nature of TFP led us to consider PCy_3, P(1-naphthyl)$_3$, **L1**, and **L2**; however, they all gave inferior results (Table 1, entries 6–9). Thus, we reasoned that a steric effect may also count. Recently, we have developed several readily available sterically hindered monophosphine ligands (**LB-Phos·HBF$_4$** and **Gorlos-Phos·HBF$_4$**, Table 1) and found nice applications in C–Cl bond activation.[6] When these ligands were tested in this reaction, interestingly, only (*Z*)-**3aa** was found, as judged by ^{1}H NMR spectroscopic analysis of the crude reaction mixture with **Gorlos-Phos·HBF$_4$** (Table 1, entry 11). Control experiments using its neutral counterpart **Gorlos-Phos** gave similar results (Table 1, entry 12), indicating

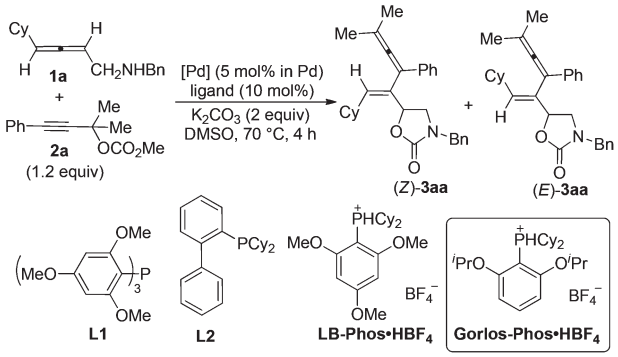
[a]*State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 345 Lingling Lu, Shanghai 200032, P. R. China*
[b]*Shanghai Key Laboratory of Green Chemistry and Chemical Process, Department of Chemistry, East China Normal University, 3663 North Zhongshan Road, Shanghai 200062, P. R. China. E-mail: masm@sioc.ac.cn; Fax: (+86)-21-6260-9305*

†Electronic supplementary information (ESI) available: ^{1}H, ^{13}C NMR and HPLC spectra of all the products. CCDC 833031 and 919207. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/c3ob40983j

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Table 1 Screening of palladium catalysts and ligands for the stereoselective formation of (*Z*)-**3aa**[a]



Entry	[Pd]	Ligand	Yield of **3aa**[b] (%)/(*Z*/*E*)
1[c]	$Pd(PPh_3)_4$	—	76 (82/18)
2	$Pd(OAc)_2$	TFP	56 (94/6)
3	$Pd(dba)_2$	TFP	83 (94/6)
4	$PdCl_2$	TFP	83 (91/9)
5	$Pd_2(dba)_3{\cdot}CHCl_3$	TFP	94 (95/5)
6	$Pd_2(dba)_3{\cdot}CHCl_3$	PCy_3	25 (84/16)
7	$Pd_2(dba)_3{\cdot}CHCl_3$	P(1-naphthyl)$_3$	37 (81/19)
8	$Pd_2(dba)_3{\cdot}CHCl_3$	**L1**	62 (92/8)
9	$Pd_2(dba)_3{\cdot}CHCl_3$	**L2**	Trace[d]
10	$Pd_2(dba)_3{\cdot}CHCl_3$	**LB-Phos·HBF$_4$**	81 (96/4)
11	**$Pd_2(dba)_3{\cdot}CHCl_3$**	**Gorlos-Phos·HBF$_4$**	**97 (>97/3)**
12	$Pd_2(dba)_3{\cdot}CHCl_3$	**Gorlos-Phos**	99 (>97/3)

[a] The reactions were carried out using **1a** (0.1 mmol), **2a** (0.12 mmol), K_2CO_3 (2 equiv.) and the indicated catalyst and ligand in DMSO (1 mL) in a Schlenk tube at 70 °C for 4 h. [b] Determined using ^{1}H NMR analysis of the crude reaction mixture. [c] No ligand was added. [d] 60% of **1a** was recovered. TFP = tri(2′-furyl)phosphine, PCy_3 = tricyclohexylphosphine, P(1-naphthyl)$_3$ = tri(1-naphthyl)phosphine.

that the cationic nature of the ligand does not affect the selectivity.

Under this set of optimal conditions, the substrate scope was examined (Table 2). To our delight, 2,3-allenyl amines containing a *c*-hexyl, *i*-Pr (Table 2, entries 1–6) or a linear alkyl substituent (Table 2, entries 7–10) at the 4-position all afforded the corresponding products (*Z*)-**3** as a single isomer in good to excellent yields. The structure and the configuration of the C=C bond of these products were confirmed unambiguously by the X-ray diffraction study of (*Z*)-**3da** (Fig. 1).[7]

Next, we examined the feasibility of extending the present system to reactions of 2,3-allenyl amines with organic halides instead of propargylic carbonates. Gratifyingly, when the reactions of 2,3-allenyl amines and phenyl iodide were carried out under the aforementioned standard conditions in the presence of a CO_2 balloon, the corresponding 5-alkenyloxazolidin-2-ones **4** were also obtained in good to excellent yields with *Z*/*E* ranging from ≥94 : 6 to >97 : 3 as determined using ^{1}H NMR analysis of the crude reaction mixture (Table 3). The structure and the configuration of the C=C bond of these products were confirmed by the X-ray diffraction study of (*Z*)-**4c** (Fig. 2).[8]

To further explore the generality of this catalyst system, the reactions of 4,5-allenoic acids **5** and propargylic carbonates **2** were surveyed and the representative results are shown in Table 4. To our delight, the reactions of 4,5-allenoic acids **5a**–**5c** with various propargylic carbonates all furnished the corresponding (*Z*)-5-(1,3,4-alkatrien-2-yl)-4,5-dihydro-2(3*H*)-furanones

Table 2 Palladium-catalyzed cyclization of 4-monosubstituted 2,3-allenyl amines with propargylic carbonates[a]



Entry	R^1	R^2, R^3, R^3	Yield of (*Z*)-**3**[b] (%)
1	Cy (**1a**)	Ph, Me, Me (**2a**)	86 (*Z*-**3aa**)
2	Cy (**1a**)	Ph, –$(CH_2)_4$– (**2b**)	86 (*Z*-**3ab**)
3	Cy (**1a**)	Me, Me, Me (**2c**)	91 (*Z*-**3ac**)
4	Cy (**1a**)	*n*-Bu, Me, Me (**2d**)	86 (*Z*-**3ad**)
5	^{i}Pr (**1b**)	Ph, Me, Me (**2a**)	85 (*Z*-**3ba**)
6	^{i}Pr (**1b**)	Ph, –$(CH_2)_5$– (**2e**)	81 (*Z*-**3be**)
7	*n*-Pr (**1c**)	Ph, Me, Me (**2a**)	82 (*Z*-**3ca**)
8	*n*-Pr (**1c**)	*n*-Bu, Me, Me (**2d**)	85 (*Z*-**3cd**)
9	Et (**1d**)	Ph, Me, Me (**2a**)	83 (*Z*-**3da**)
10	Et (**1d**)	Ph, –$(CH_2)_5$– (**2e**)	84 (*Z*-**3de**)

[a] The reactions were carried out on a 0.3 mmol scale of **1** in 3 mL of DMSO in a Schlenk tube. [b] Isolated yields of (*Z*)-**3**.

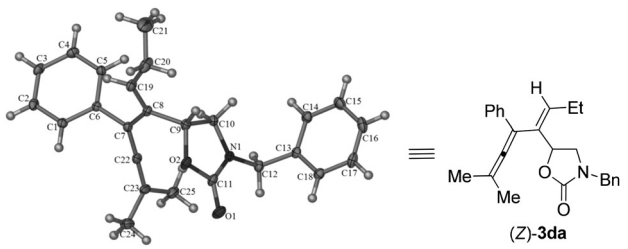
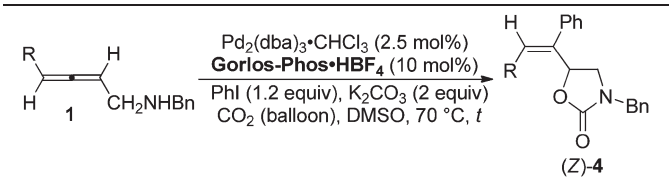


Fig. 1 ORTEP representation of (*Z*)-**3da**.

Table 3 Stereoselective synthesis of (*Z*)-5-alkenyloxazolidin-2-ones from 4-monosubstituted 2,3-allenyl amines, phenyl iodide and carbon dioxide[a]



Entry	R^1	*t* (h)	*Z*/*E*[b]	Yield of (*Z*)-**4**[c] (%)
1	*c*-C_6H_{11} (**1a**)	8	≥96/4	87 (*Z*-**4a**)
2	*n*-Pr (**1c**)	6	≥94/6	82 (*Z*-**4c**)
3	*n*-Bu (**1e**)	6	≥94/6	80 (*Z*-**4e**)
4	^{t}Bu (**1f**)	7	>97/3	92 (*Z*-**4f**)
5	*n*-C_7H_{15} (**1g**)	6	≥96/4	84 (*Z*-**4g**)

[a] The reactions were carried out on a 0.1 mmol scale of **1** in 1 mL of DMSO in a Schlenk tube with a CO_2 balloon. [b] Determined using ^{1}H NMR analysis of the crude reaction mixture. [c] Isolated yields of (*Z*)-**4**.

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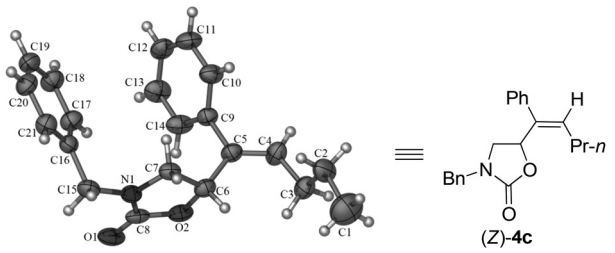


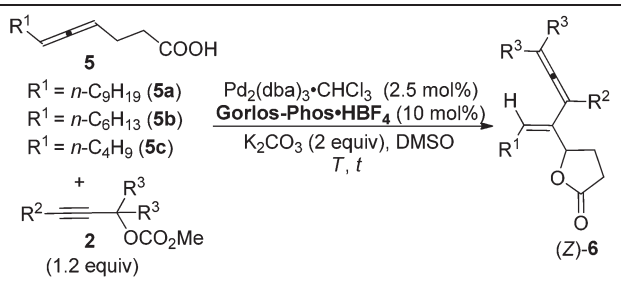
Fig. 2 ORTEP representation of (*Z*)-**4c**.

6 in good yields with high stereoselectivity ($Z/E \geq 95:5$) (Table 4).

Having achieved the stereoselective synthesis of (*Z*)-5-(1,3,4-alkatrien-2-yl)oxazolidin-2-ones **3**, we wondered whether their chiral counterparts could be obtained by employing the axially chiral 2,3-allenyl amines[9] *via* an axial-to-central chirality transfer process. However, when enantioenriched 2,3-allenyl amine (*S*)-**1a** (98% ee) was subjected to the standard conditions, the corresponding product (*R*,*Z*)-**3aa** was obtained with only 80% ee, indicating a serious racemization (Table S1, entry 1, in the ESI†). Gratifyingly, after extensive study, we eventually found that the reaction temperature had a significant influence on the chirality transfer process (Table S1, entries 2–4†), and enantioenriched 5-(1,3,4-alkatrien-2-yl)oxazolidin-2-one (*R*,*Z*)-**3** can be accessed in moderate to good yields with only a slight decrease in enantiomeric purity at lower reaction temperatures ranging from 45 to 60 °C (Table 5, entries 1 and 3–6). The reaction of 2,3-allenyl amine (*S*)-**1a** with cyclic propargylic carbonate **2b** was carried out at 70 °C for the full conversion of (*S*)-**1a** (Table 5, entry 2).

A possible mechanism involving the two different types of intermediates shown in Scheme 2 was proposed.[4d] Path a

Table 4 Stereoselective synthesis of (*Z*)-5-(1,3,4-alkatrien-2-yl)-4,5-dihydro-2-(3*H*)-furanones from 4,5-allenoic acids and propargylic carbonates[a]



Entry	5	R^2, R^3, R^3	T (°C) (t (h))	Z/E^b	Yield of (*Z*)-**6**^{c} (%)
1	**5a**	Ph, Me, Me (**2a**)	70 (3)	≥97/3	86^d (*Z*-**6aa**)
2	**5a**	Ph, –$(CH_2)_4$– (**2b**)	60 (7)	≥98/2	85 (*Z*-**6ab**)
3	**5a**	4-$O_2NC_6H_4$, Me, Me (**2f**)	40 (8)	≥96/4	80 (*Z*-**6af**)
4	**5b**	Ph, Me, Me (**2a**)	60 (6)	≥95/5	84 (*Z*-**6ba**)
5	**5c**	Ph, Me, Me (**2a**)	60 (6)	≥96/4	80 (*Z*-**6ca**)

[a] The reactions were carried out on a 0.3 mmol scale of **5** in 3 mL of DMSO in a Schlenk tube. [b] Determined using ^{1}H NMR analysis of the crude reaction mixture. [c] Isolated yields of (*Z*)-**6**. [d] Containing minor (*ca.* 2%) *E* isomer.

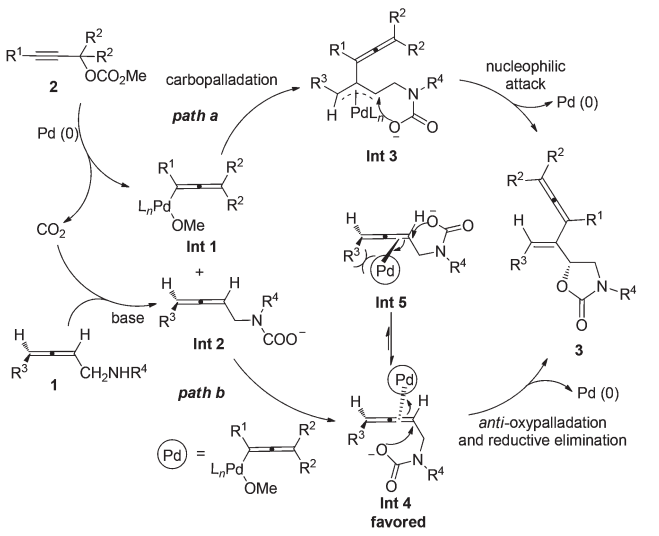
Table 5 Synthesis of optically active 5-(1,3,4-alkatrien-2-yl)oxazolidin-2-ones from axially chiral 2,3-allenyl amines[a]



Entry	(*S*)-**1**	**2** R^2, R^3, R^3	T (°C) (t (h))	Yield of (*R*,*Z*)-**3**^{b} (%)
1	**1a**	Ph, Me, Me (**2a**)	55 (3)	82 (**3aa**, 95% ee)
2	**1a**	Ph, –$(CH_2)_4$– (**2b**)	70 (2)	68 (**3ab**, 94% ee)
3	**1a**	Me, Me, Me (**2c**)	50 (6)	83 (**3ac**, 96% ee)
4	**1a**	*n*-Bu, Me, Me (**2d**)	50 (4)	84 (**3ad**, 94% ee)
5	**1c**	Ph, Me, Me (**2a**)	60 (3)	68 (**3ca**, 95% ee)
6	**1c**	*n*-Bu, Me, Me (**2d**)	45 (4)	66 (**3cd**, 95% ee)

[a] The reactions were carried out on a 0.1 mmol scale of (*S*)-**1** in 1 mL of DMSO in a Schlenk tube. [b] Isolated yields of (*R*,*Z*)-**3**.

involves the formation of a π-allyl palladium species **Int 3** *via* carbopalladation whereas path b affords the product **3** *via anti*-oxypalladation of **Int 4** followed by reductive elimination. As the enantioenriched 2,3-allenyl amines are capable of transferring the axial chirality into the newly formed stereogenic center of 5-(1,3,4-alkatrien-2-yl)oxazolidin-2-ones without obvious loss of enantiomeric purity, we are in favor of path b since racemization may occur through a σ–π–σ process if the π-allyl palladium species **Int 3** is involved *via* path a.[3,10] The steric bulky nature of the ligand **Gorlos-Phos** helps to increase the steric difference between **Int 4** and **Int 5**, making the formation of **Int 4** more favorable. Therefore, *Z* isomers were obtained *via* **Int 4** (path b). The absolute configurations of the enantioenriched 5-(1,3,4-alkatrien-2-yl)oxazolidin-2-ones were



Scheme 2 A possible mechanism.

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tentatively assigned as *R* based on this mechanism involving *anti*-oxypalladation.[3,10]

Conclusions

In conclusion, we have developed an efficient and highly stereoselective protocol for the construction of synthetically attractive oxazolidin-2-one[11] and γ-butyrolactone[12] scaffolds with the challenging issue of *Z*/*E* stereoselectivity being nicely addressed by taking advantage of the readily accessible, sterically hindered monophosphine ligand **Gorlos-Phos·HBF$_4$**. The simplicity, high stereocontrolling ability as well as promising generality of this catalyst system clearly demonstrated its vast potential for stereoselective synthesis in palladium-catalyzed *exo*-mode cyclization reactions of allenes bearing a nucleophilic functionality with organic halides or their equivalents. Further studies are aimed at achieving a better understanding of the mechanism and applying such a catalyst system in related transformations.

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7 Crystal data for compound (*Z*)-**3da**: $C_{25}H_{27}NO_2$, MW = 373.48, tetragonal, space group $I4(1)/a$, final *R* indices [$I > 2\sigma(I)$], $R_1 = 0.0447$, w$R_2 = 0.0868$; *R* indices (all data): $R_1 = 0.0705$, w$R_2 = 0.0966$; $a = 38.342(3)$ Å, $b = 38.342(3)$ Å, $c = 5.7826(4)$ Å, $\alpha = 90°$, $\beta = 90°$, $\gamma = 90°$, $V = 8501.1(10)$ Å^3, $T = 173(2)$ K, $Z = 16$, reflections collected/unique: 46 864/3736 ($R_{int} = 0.0836$), the number of observations [$>2\sigma(I)$] 2736, parameters: 253. CCDC 833031 contains the supplementary crystallographic data for this paper.

8 Crystal data for compound (*Z*)-**4c**: $C_{21}H_{23}NO_2$, MW = 321.10, monoclinic, space group $P2(1)/c$, final *R* indices [$I > 2\sigma(I)$], $R_1 = 0.0685$, w$R_2 = 0.2050$; *R* indices (all data), $R_1 = 0.1043$, w$R_2 = 0.2462$; $a = 13.9084(18)$ Å, $b = 5.9803(8)$ Å, $c = 22.981(3)$ Å, $\alpha = 90°$, $\beta = 105.356(4)°$, $\gamma = 90°$, $V = 1843.3(4)$ Å^3, $T = 296(2)$ K, $Z = 4$, reflections collected/unique 20 338/3223 ($R_{int} = 0.0359$), the number of observations [$>2\sigma(I)$] 1918, parameters: 254. CCDC 919207 contains the supplementary crystallographic data for this paper.

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